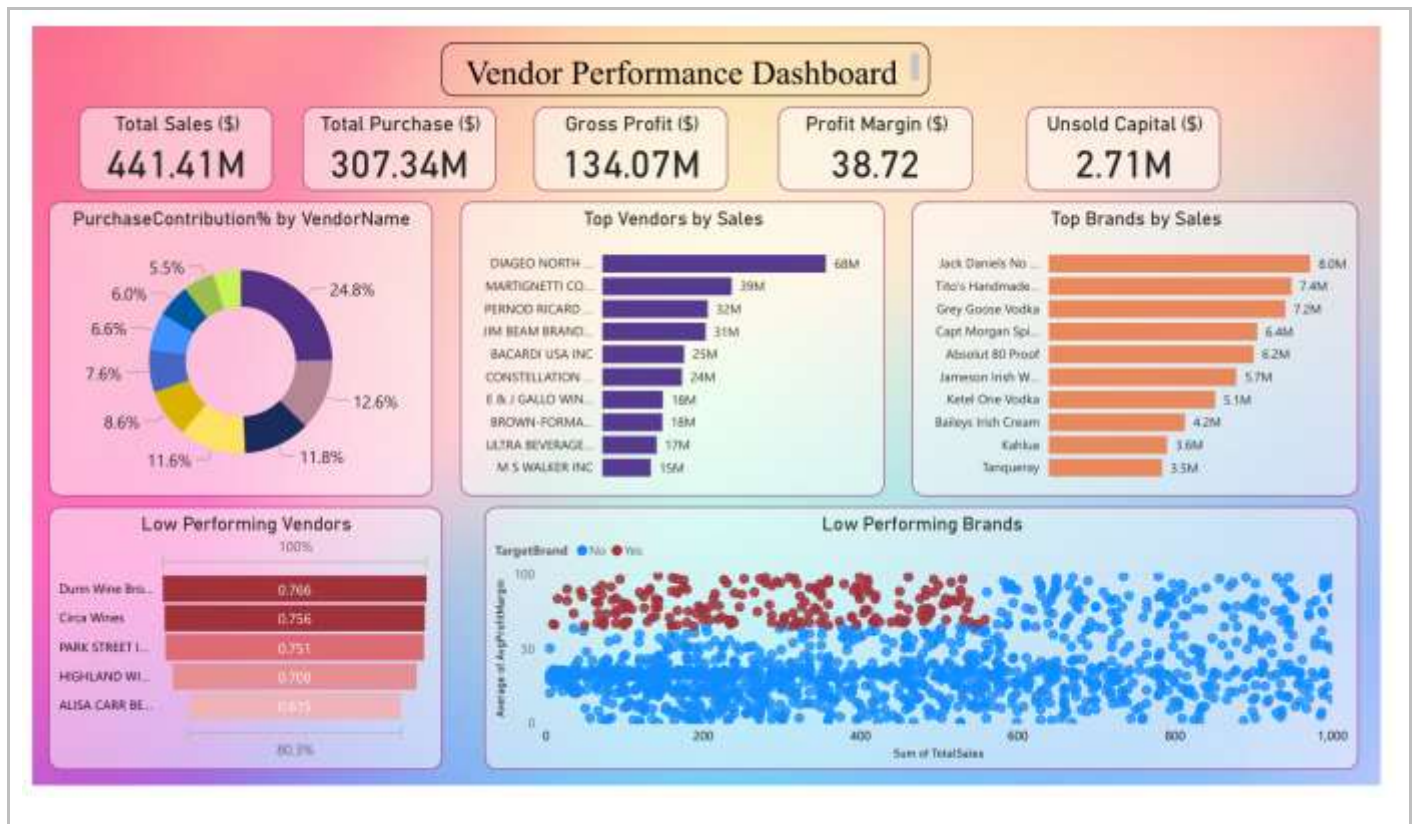


VENDOR PERFORMANCE ANALYSIS

Business Intelligence & Data Analytics Report

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Tools: Python • SQL • Power BI



Business Problem

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. The business needs to avoid losses caused by inefficient pricing, poor inventory turnover, and excessive dependency on a limited number of vendors.

The analysis aims to:

Identify underperforming brands that require promotional or pricing adjustments.

Determine top vendors contributing to sales and gross profit.

Analyze the impact of bulk purchasing on unit costs.

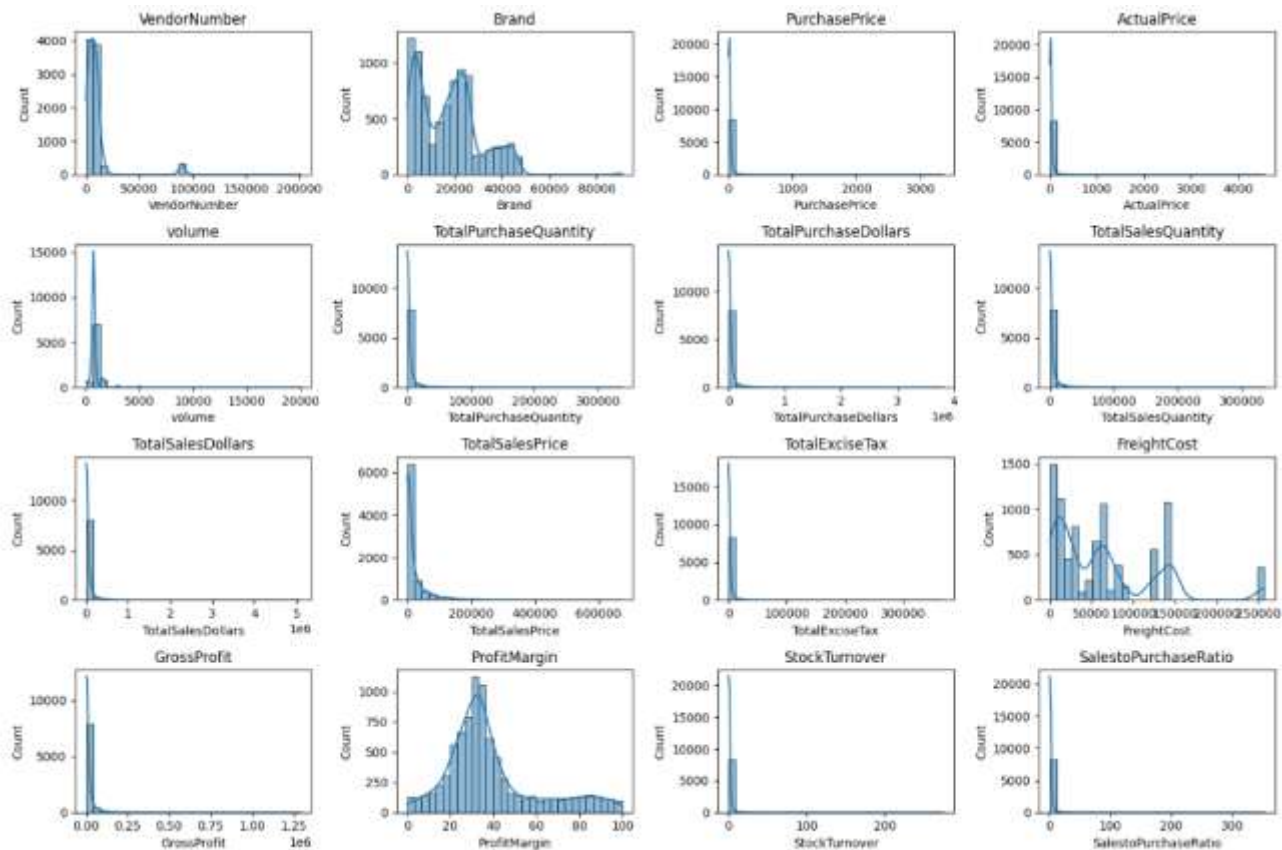
Assess inventory turnover to reduce holding costs and improve efficiency.

Investigate profitability differences between high-performing and low-performing vendors.

Exploratory Data Analysis

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalestoPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02



Gross Profit: The minimum gross profit is **-\$52,002.78**, indicating potential losses due to high costs or heavy discounts. This could be due to products being sold at prices lower than their purchase costs.

Profit Margin: The minimum value is **$-\infty$** , which suggests instances where revenue is zero, resulting in undefined or extremely negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show **zero sales**, indicating that they were purchased but never sold. These may represent slow-moving or obsolete inventory, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values of **\$5,681.81** and **\$7,499.99** are significantly higher than the mean values of **\$24.39** and **\$35.64**, respectively, indicating substantial price variation and the presence of potentially premium-priced products.

Freight Cost: Freight costs show extreme variation, ranging from **\$0.09** to **\$257,032.07**. This may indicate differences in logistics requirements, bulk shipments, or unusually high shipping costs across different products.

Stock Turnover: Stock turnover ranges from **0 to 274.5**, suggesting substantial variation in inventory movement. Some products may sell rapidly, while others remain in inventory for extended periods. A value greater than **1** indicates that sales exceed the purchased quantity during the analyzed period, potentially because existing/older inventory was used to fulfill sales.

Data Filtering

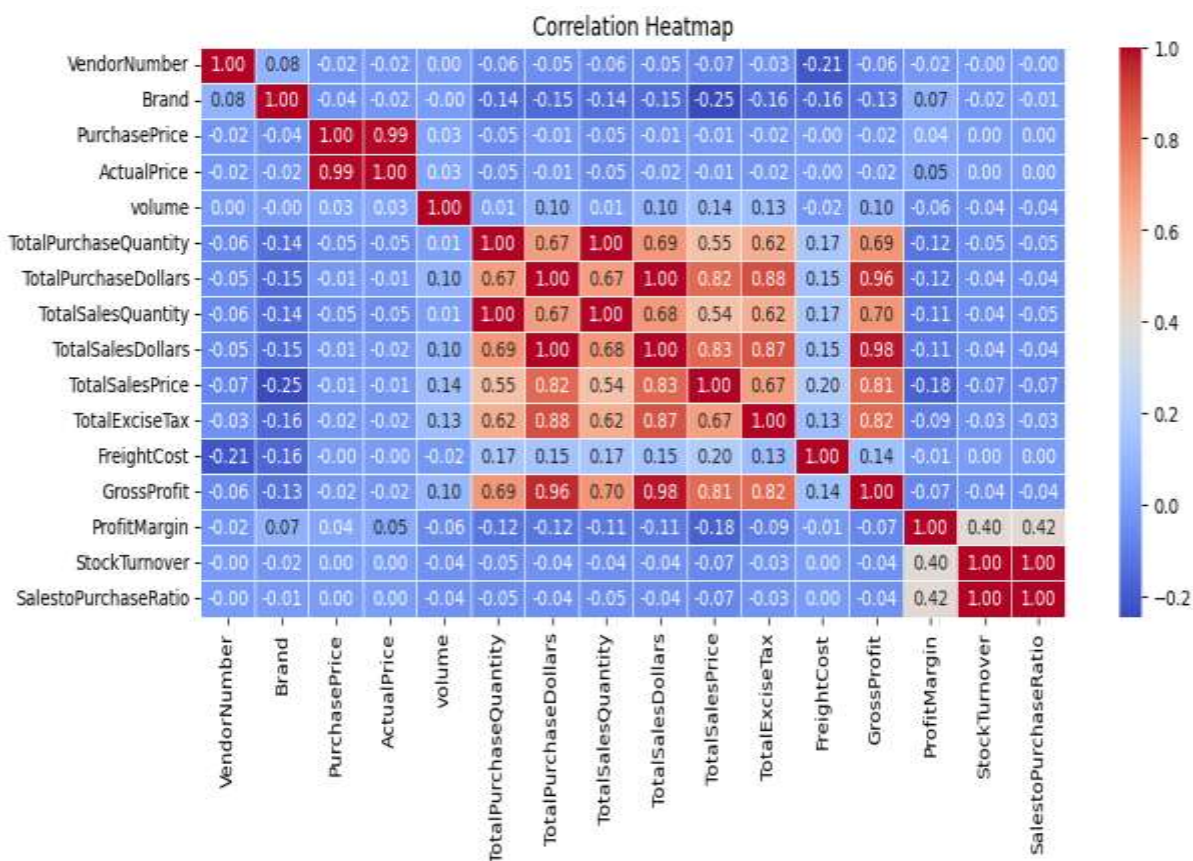
To improve the reliability of the business analysis, inconsistent observations were removed using the following criteria:

Gross Profit ≤ 0 — exclude transactions leading to losses.

Profit Margin ≤ 0 — focus the comparative analysis on profitable transactions.

Total Sales Quantity = 0 — exclude inventory that was purchased but never sold.

Correlation Analysis



Correlation Insights

- **Purchase Price vs. Total Sales Dollars & Gross Profit:** Weak negative correlations of **-0.012** and **-0.016**, respectively, indicating that variations in purchase price do not have a significant linear relationship with sales revenue or gross profit.
- **Total Purchase Quantity vs. Total Sales Quantity:** A very strong positive correlation of **0.999**, indicating that purchased quantities and sales quantities move almost proportionally in the dataset.
- **Profit Margin vs. Total Sales Price:** A weak negative correlation of **-0.179**, suggesting that higher sales prices are associated with slightly lower profit margins. This may reflect competitive pricing pressure or differences in product-level economics.
- **Stock Turnover vs. Gross Profit & Profit Margin:** Weak negative correlations of **-0.038** and **-0.055**, respectively, indicating that faster inventory turnover does **not necessarily** correspond to higher gross profit or profit margins.

Research Questions

- ✚ Which brands have low sales but high profit margins and may benefit from targeted promotion or price optimization?

Brands with low sales but high profit margins:

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

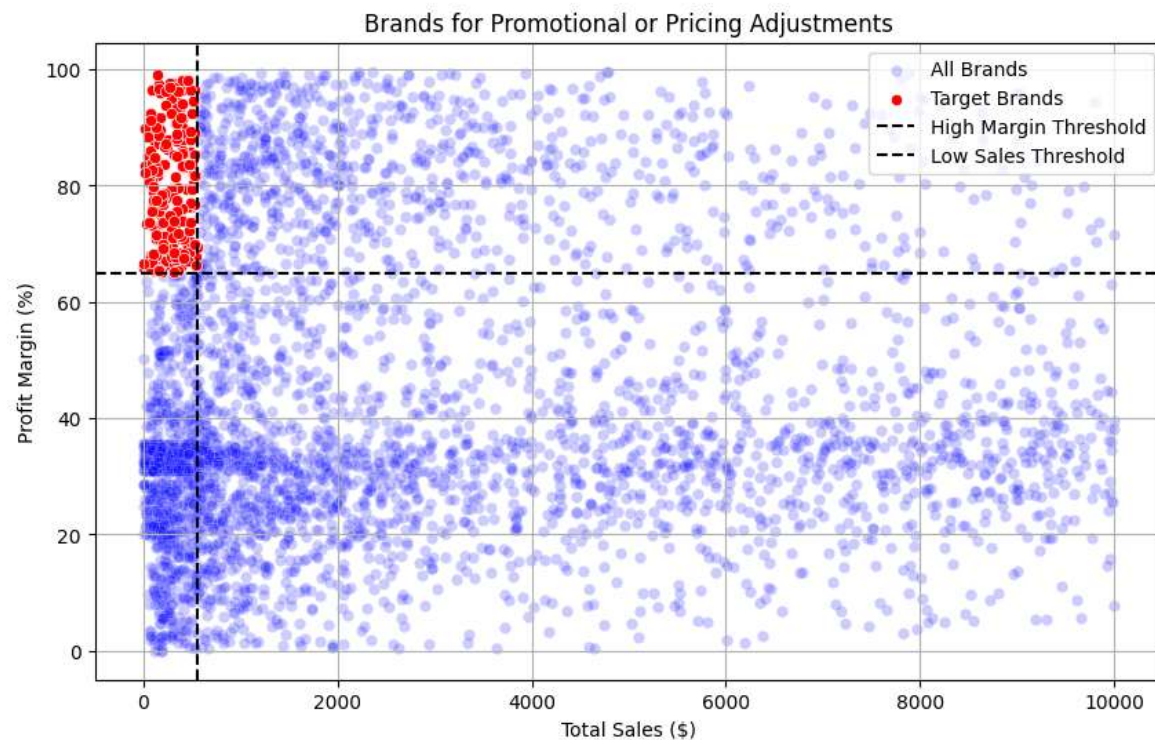
198 rows × 3 columns

198 brands exhibit **lower sales but higher profit margins**, indicating that these brands have strong profitability potential but relatively weak sales volume.

These brands could benefit from:

- **Targeted marketing campaigns** to increase product visibility and customer demand.
- **Promotional offers** to stimulate sales volume while maintaining acceptable margins.
- **Price optimization** to find a balance between sales volume and profitability.
- **Improved distribution** to increase product availability and market reach.

Brands for Promotional or Pricing Adjustments



Key Business Insight

The objective is to **increase sales volume without significantly compromising profit margins**. These brands represent potential growth opportunities because their existing high margins provide some flexibility for strategic pricing and promotional initiatives.

Visualization

Figure: Brands for Promotional or Pricing Adjustments

The scatter plot identifies the target brands based on **low sales** and **high profit margins**. The dashed lines represent the selected thresholds for identifying these brands.

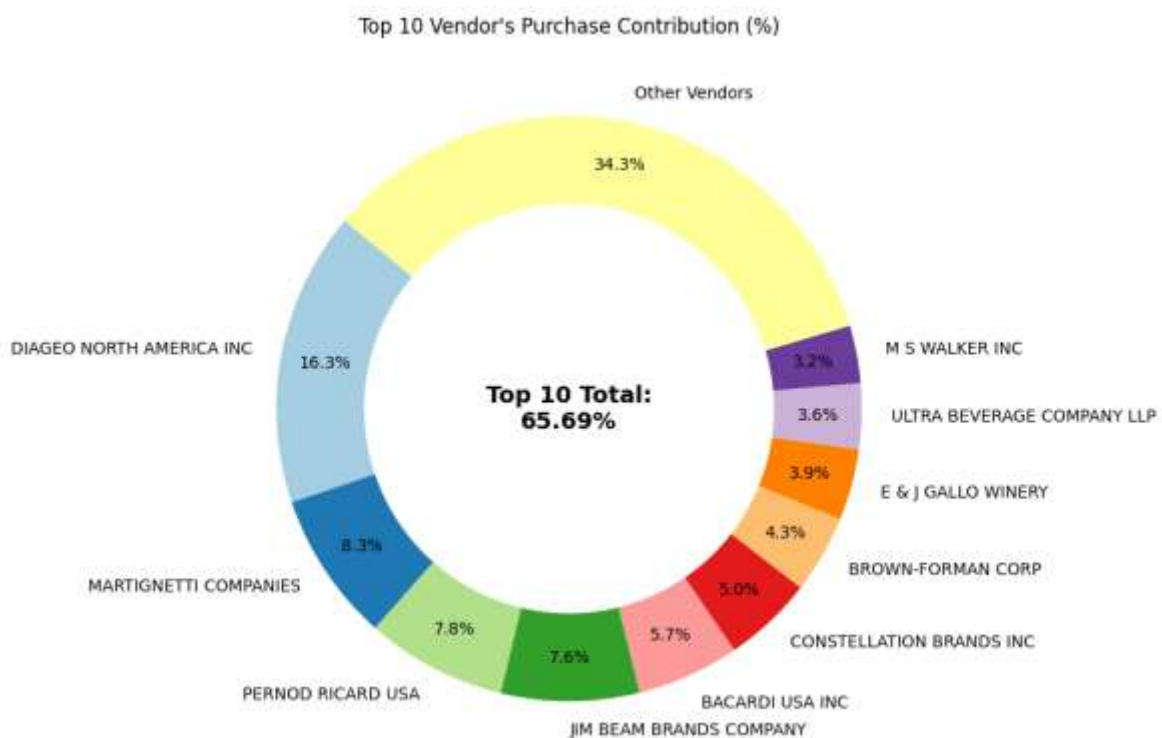
Key Result:

198 brands fall into the low-sales/high-margin segment and should be considered for targeted

promotional or pricing strategies.

What proportion of purchases is concentrated among the top vendors?

The top 10 vendors account for 65.69% of total purchases, while the remaining vendors account for 34.31%. This concentration indicates dependence on a relatively small group of suppliers and creates potential supply-chain risk if one or more major vendors experience disruption.



Impact of Bulk Purchasing on Cost Savings

Unit purchase price declines substantially as order size increases. The large-order unit price is approximately 72% lower than the small-order unit price based on these reported averages. This supports the use of bulk purchasing where demand, storage capacity, cash flow, and inventory risk justify larger orders.

	UnitPurchasePrice
OrderSize	
Large	39.068186
Medium	15.486414
small	10.777625

- Which vendors have low inventory turnover and high unsold inventory value?

Total Unsold Inventory Capital: \$2.71M

Slow-moving inventory ties up working capital, increases storage costs, reduces cash-flow efficiency, and can negatively affect overall profitability.

Identifying vendors with **low inventory turnover** helps the business improve stock management, reduce excess inventory, and minimize financial strain.

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

Profit Margin Comparison: High vs. Low-Performing Vendors

Profit Margin Comparison

The analysis compares the profit margins of **top-performing vendors** with those of **low-performing vendors** using 95% confidence intervals.

- **Top Vendors:** 95% CI = **30.74%–31.61%**; Mean Profit Margin = **31.17%**
- **Low-Performing Vendors:** 95% CI = **40.48%–42.62%**; Mean Profit Margin = **41.55%**

The results show that **low-performing vendors have substantially higher profit margins than top-performing vendors**, despite generating lower sales volumes. This suggests that low sales performance is not necessarily caused by poor margins. Instead, factors such as **pricing strategy, limited market reach, weaker demand, marketing effectiveness, or distribution constraints** may be contributing to their lower sales.

Actionable Insights

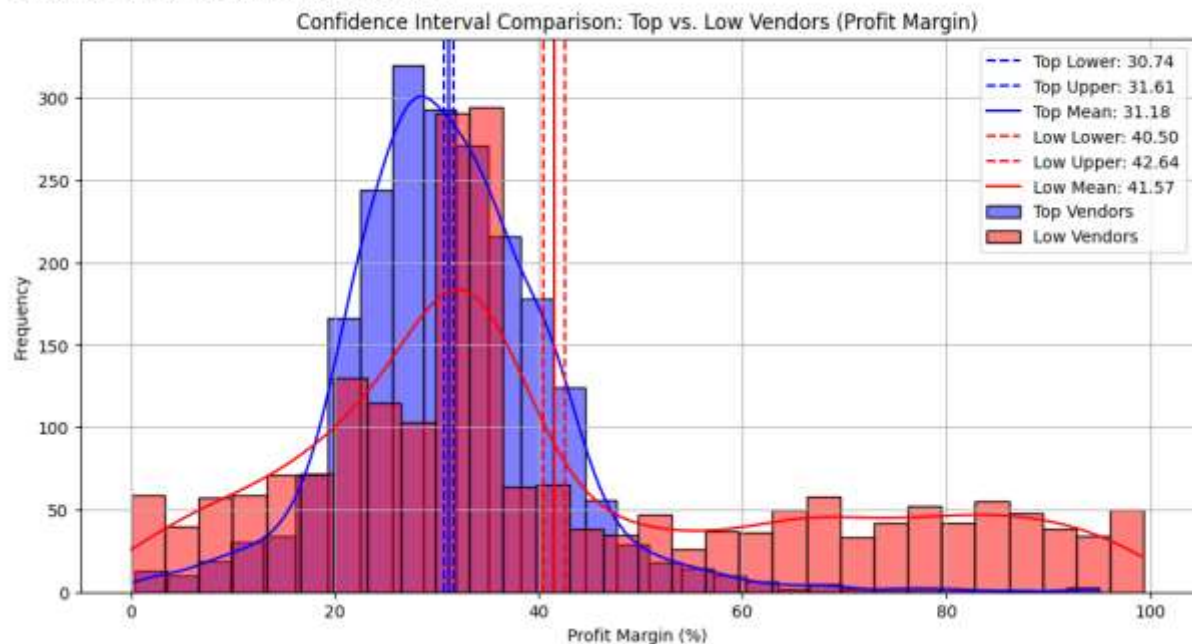
Top-performing vendors:

- Optimize pricing to balance revenue growth and profitability.
- Identify opportunities to reduce operational and supply-chain costs.
- Use bundled offers or targeted promotions to increase sales while protecting margins.
- Maintain strong inventory availability for high-demand products.

Low-performing vendors:

- Improve marketing and promotional activities to increase product awareness.
- Review pricing strategies to determine whether margins can be selectively reduced to stimulate demand.
- Expand or improve distribution channels.
- Identify products with high margins but low demand and develop targeted sales strategies.

Top Vendors 95% CI: (30.74, 31.61), Mean: 31.18
Low Vendors 95% CI: (40.50, 42.64), Mean: 41.57



Statistical Validation of Profit Margin Differences

Hypothesis Testing:

To statistically validate whether the observed difference in profit margins between top-performing and low-performing vendors is meaningful, a hypothesis test was conducted.

- **H₀ (Null Hypothesis):**
There is **no statistically significant difference** in profit margins between top-performing and low-performing vendors.
- **H₁ (Alternative Hypothesis):**
There is a **statistically significant difference** in profit margins between top-performing and low-performing vendors.

Result:

The **null hypothesis (H₀) is rejected**, indicating that the difference in profit margins between the two vendor groups is statistically significant.

This provides statistical evidence that the two groups have **distinct profitability characteristics**, rather than the observed difference being attributable solely to random variation.

Business Implication

The findings suggest that **high-margin vendors and top-selling vendors may require different business strategies:**

- **High-margin, low-performing vendors:** Focus on improving market reach, marketing, pricing, and sales volume while preserving their relatively strong margins.
- **Top-selling vendors:** Focus on cost efficiency, operational optimization, and maintaining healthy margins while sustaining high sales volumes.

Final Recommendations

- **Re-evaluate pricing for low-sales, high-margin brands** to increase sales volume without significantly compromising profitability.
- **Diversify vendor partnerships** to reduce dependency on a small number of suppliers and mitigate potential supply-chain risks.
- **Leverage bulk purchasing advantages** to maintain competitive pricing while optimizing inventory management and procurement costs.
- **Optimize slow-moving inventory** by adjusting purchase quantities, launching clearance sales, and revising inventory and storage strategies.
- **Enhance marketing and distribution strategies for low-performing vendors** to increase sales volume while maintaining healthy profit margins.
- **Implement targeted pricing and promotional strategies** for brands with high profit margins but low sales to improve market reach and demand.
- **Focus on cost efficiency for top-selling vendors** by reducing unnecessary operational costs and optimizing procurement and logistics.
- **Use inventory turnover as a performance indicator** to identify vendors and products that require inventory-management interventions.
- **Monitor vendor concentration** and gradually diversify purchasing across reliable suppliers to improve supply-chain resilience.

Expected Business Impact:

By implementing these recommendations, the company can increase sustainable profitability, reduce excess inventory, improve cash-flow efficiency, mitigate supplier concentration risks, and enhance overall operational efficiency.

Project Learnings

- Performed data cleaning and transformation on vendor and sales data.
- Used Python and Pandas for data preparation, aggregation, and exploratory analysis.
- Used Matplotlib and Seaborn to visualize distributions, relationships, and vendor segments.
- Applied correlation analysis to identify relationships among purchasing, sales, pricing, profitability, and inventory metrics.
- Learned how to identify and interpret negative values, zero-sales observations, and extreme outliers.
- Used business rules to filter inconsistent observations before comparative analysis.
- Applied confidence intervals and hypothesis testing to validate differences between vendor groups.
- Translated analytical findings into recommendations related to pricing, procurement, inventory, vendor diversification, and marketing.
- Developed a business-oriented view of vendor performance rather than relying only on descriptive statistics.

Conclusion

The Vendor Performance Analysis provides a structured view of how purchasing behavior, sales volume, pricing, profitability, inventory turnover, and vendor concentration affect operational performance. The analysis identifies 198 low-sales/high-margin brands as potential targets for marketing or pricing actions, shows that the top 10 vendors account for 65.69% of purchases, demonstrates a substantial unit-cost advantage from large orders, and highlights approximately \$2.71 million in unsold inventory capital.

The comparison between vendor groups further shows a higher average profit margin among low-performing vendors (41.55%) than top-performing vendors (31.17%), and hypothesis

testing rejects the null hypothesis of no difference. Together, these findings suggest that the business should balance growth and profitability through targeted pricing, vendor diversification, selective bulk purchasing, inventory optimization, and stronger marketing and distribution strategies.

Implementing these recommendations can support sustainable profitability, reduce supply-chain and inventory risk, improve working-capital efficiency, and strengthen overall operational decision-making.